

EDITORIAL

The brain—from neurodevelopment to neurodegeneration

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The brain is the organ that orchestrates the whole body. To do so, neurons and glia in different brain regions have evolved to be highly specialized and are regulated by diverse receptors and neuro- and gliotransmitters. Some of the features of the brain, including a protective brain–blood barrier, high-energy demand, and electrical activity, mean that neurons and astrocytes are uniquely diverse components of tissue that vary greatly even across brain regions. This Special Issue features 19 review articles that cover different aspects of the broad field of neuroscience—from neurodevelopment, physiology and memory formation to the mechanisms underlying the development of neurodegenerative diseases.

The brain is a sophisticated network of cells that, given its complexity, remains poorly understood despite decades of research. Most of the known human neurological disorders, including neurodegenerative diseases, are presently incurable [1,2]. This Special Issue of *The FEBS Journal* provides the reader with a broad view of various angles of neuroscience—activation of neurons with receptors, excitation and inhibition balance, memory formation, neurodevelopment and, importantly, a modern opinion on the development of different pathologies of the central nervous system (CNS).

Neurons are responsible for calculation of the needs of the body throughout various environmental conditions and triggers. The bi-directional communication between the CNS and the body—interoception—helps to control multiple processes, including the driving force to eat, which is controlled by neurons in the hypothalamic arcuate nucleus that express agouti-related

peptide (AgRP). In this Special Issue, Deem et al. [3] discuss key recent findings that improve our understanding of the AgRP neuron population.

Neuronal function in the different brain regions is regulated by multiple receptors and neurotransmitters. However, could the function of the whole brain be modulated by a single receptor type? Jancke et al. [4] seek to bridge the gap between single receptor-type activity and whole-brain dynamics, taking as a model the impact of single serotonergic (5-HT) receptor types on cortical dynamics. GABAergic interneurons are a neuronal cell type that is responsible for the balance of excitation and inhibition in the brain, and Yang et al. [5] highlight recent findings on the development of cortical interneuron and interneuron dysfunction, with implications for understanding certain neurodevelopmental disorders.

Hippocampal circuits are crucial for cognitive performance, and changes in this sophisticated network

Abbreviations

AgRP, agouti-related peptide; AMPK, adenosine monophosphate-activated protein kinase; ASD, autism spectrum disorder; CNS, central nervous system; LTD, long-term depression; LTP, long time potentiation; PARP-1, poly (ADP-ribose)-polymerase-1.

may lead to neurological disease [6]. Robledo-Menendez et al. [6] provide an overview of how cannabinoid receptors control hippocampal functions in physiology and pathology. The hippocampus is also the key brain region responsible for memory formation, a process in which long-term potentiation (LTP) and long-term depression (LTD)—forms of synaptic plasticity—are believed to play major roles. Bin Ibrahim et al. [7] review the modern view on the different forms of synaptic plasticity and its regulatory mechanisms in the hippocampus. Synaptic plasticity has been shown to be dependent on neuron–glia interactions and Wang et al. [8] highlight the instructive roles of astrocytes, the most common glial cells, in hippocampal synaptic plasticity through modulation of the secretion of gliotransmitters and synaptogenic proteins or via contact-mediated signaling pathways. Also within the theme of neuron–glia interactions, Ifran et al. [9] describe how microglia phagocytose oligodendrocyte progenitor cells during postnatal development and the role of microglial phagocytosis in white versus gray matter maturation.

Another brain region—the striatum—controls a number of cellular functions, including motor control, learning, motor skills, cognitive function, and emotion. Cataldi et al. [10] describe the role of the striatum and its connections in motor learning and performance and how these can be impaired in pathological conditions such as Parkinson's disease.

Brain cells—neurons and glia—maintain homeostasis and regulate their function using a general cellular mechanism and some cell-specific proteins. Endocytosis is an essential cellular process required for multiple physiological functions in various cell types, and in this Special Issue, the specificity of endocytosis for brain cell function is summarized [11]. Maintenance of energy metabolism is also vitally important for cells in the brain, which consumes 10 times more glucose and oxygen than other tissues [12], rendering the tight regulation of energy metabolism particularly crucial. The adenosine monophosphate-activated protein kinase (AMPK) is a metabolic sensor that maintains energy balance, and Muraleedharan and Dasgupta review the important role of this enzyme in brain glucose metabolism. Unsurprisingly, AMPK dysregulation is implicated in several neurodegenerative diseases [13].

Due to their morphological structure, neurons transport molecules along microtubules to the axons and dendrites for a relatively long distance and kinesin superfamily proteins play a role of molecular motors. Fan and Lay [14] summarize the major kinesin motors and their associated cargoes in the physiology of neurons.

Despite the protective wall provided by the blood–brain barrier, brain cells can be infected by diverse viruses. Nath and Jonson [15] provide an overview of the factors that lead to viral persistence in the brain, how viruses penetrate the brain and how these pathogens evade the antiviral factors of the immune system.

Changes in brain function lead to a host of neurological diseases, and several review articles in this Issue focus on the molecular mechanisms underpinning brain pathologies. Autism spectrum disorder (ASD) is a common neurological disorder that is associated with altered brain connectivity [16]. Lu and Hsueh [16] discuss recent advances in the study of the mechanisms underlying ASD and highlight protein synthesis as a modifiable target for autism-related pathologies. A role for DNA damage and activation of DNA repair enzymes is recognizable in the mechanism of neurodegenerative disorders such as Parkinson's disease and Alzheimer's disease [17]. Also in this Issue, Mao and Zhang [17] describe the role of the DNA repair enzyme poly (ADP-ribose)-polymerase-1 (PARP-1) in neurodegenerative diseases and aging. One of the major histopathological hallmarks of Alzheimer's disease is the extracellular senile plaque, which consists of aggregated β -amyloid. β -amyloid is toxic for neurons and pinpointing the mechanism of this neurotoxicity may help us to understand the development of Alzheimer's disease. Konstantoulea et al. [18] discuss the heterotypic interactions that occur between amyloids and other proteins and macromolecules and how these contribute to disease. Moreover, Ginsberg et al. [19] show the development of pathology of Alzheimer's disease through the links of stressor-induced protein interactome network perturbations to the formation of pathologic scaffolds termed epichaperomes. The role of myeloid cells in retinal inflammation and degeneration and brain neurodegeneration is discussed in the review by Guo et al. [20].

Tight regulation of the innate immune system is critical for normal CNS development and function. Accordingly, aberrations that lead to neuroinflammation have been shown to be involved in a number of neurodegenerative diseases [21]. Fatoba et al. [21] describe how the complement system contributes to normal brain development and function and reacts to pathological insults in complement-mediated neuroinflammatory responses leading to neurodegeneration. Normal neurodevelopment is crucial for healthy brain function during whole life and any changes in this process lead to brain dysfunction and disease. Neurodevelopmental disorders are induced by mutations in chromatin regulators. Bonifas and Iwase [22] explore soma-to-germline transformation in chromatin-linked neurodevelopmental disorders.

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